

EQ Prerequisite: Minimum Indoor Air Quality Performance

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To contribute to the comfort and well-being of all building occupants by establishing minimum standards for indoor air quality (IAQ).

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Mechanically Ventilated Spaces

For mechanically ventilated spaces, meet the requirements of ASHRAE Standard 62.1-2016, Sections 4, 5, 6.2, 6.5, and 7, or a local equivalent, whichever is more stringent.

AND

Provide outdoor air monitors for all mechanical ventilation systems with outdoor air intake flow greater than 1000 cfm (472 L/s). The monitoring device must be capable of measuring the minimum outdoor air intake flow and be capable of measuring the design minimum outdoor air intake flow with an accuracy of +/-10%. An alarm must indicate when the outdoor airflow value varies by 15% or more from the setpoint.

Alternatively, for constant-volume systems that do not employ demand control ventilation, provide an indicator capable of confirming the intake damper is open to the position needed to maintain the design minimum outdoor airflow as determined during the system startup and balancing.

Naturally Ventilated Spaces

For naturally ventilated spaces, meet one of the following ventilation requirements.

- ▶ **Option 1. ASHRAE prescriptive natural ventilation compliance path**
Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.1, 6.4.3, 6.4.4, and 6.5.

OR

- ▶ **Option 2. ASHRAE Engineered natural ventilation system compliance path**
Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.2, 6.4.3, 6.4.4 and 6.5

OR

► **Option 3. Historic building using ASHRAE prescriptive natural ventilation path**

This option is available to projects located in a building registered as a local or national historic building.

Meet the requirements of ASHRAE 62.1-2016 with addendum I, Sections 4, 6.4.1.1-6.4.1.6, 6.4.3, 6.4.4, and 6.5.

AND

Comply with at least one of the following monitoring strategies.

- Provide a direct exhaust airflow measurement device capable of measuring the exhaust airflow. This device must measure the exhaust airflow with an accuracy of +/-10% of the design minimum exhaust airflow rate. An alarm must indicate when airflow values vary by 15% or more from the exhaust airflow setpoint. *This strategy is not allowed for projects using Natural Ventilation Option 3. Historic building.*
- Provide automatic indication devices on all natural ventilation openings intended to meet the minimum opening requirements. An alarm must indicate when any one of the openings is closed during occupied hours.
- Monitor carbon dioxide (CO₂) concentrations within each thermal zone. CO₂ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor and within the thermal zone. CO₂ monitors must have an audible or visual indicator or alert the building automation system if the sensed CO₂ concentration exceeds the setpoint by more than 10%. Calculate appropriate CO₂ setpoints using the methods in ASHRAE 62.1-2016, Appendix D.

All Spaces

The indoor air quality procedure defined in ASHRAE Standard 62.1-2016, Section 6.3 may not be used to comply with this prerequisite.

CS only

Mechanical ventilation systems must be capable of meeting projected ventilation levels and monitoring based on the requirements of anticipated future tenants.

Residential only

In addition to the requirements above, if the project building contains residential units, each dwelling unit must meet all of the following requirements:

- Design and install a dwelling-unit mechanical ventilation system that complies with ASHRAE 62.2-2016, Sections 4, 6.7, and 6.8, or a local equivalent, whichever is more stringent. Supply and balanced mechanical ventilation systems must be designed and constructed to provide ventilation air directly from the outdoors. Mechanical ventilation systems are not required when the project meets the exception to Section 4.1.1 of ASHRAE 62.2-2016.
- Design and install local mechanical exhaust systems in each kitchen and bathroom (including half-baths) that comply with ASHRAE 62.2-2016, Sections 5 and 7 or local equivalent, whichever is more stringent. Exhaust air to the outdoors. Do not route exhaust ducts to terminate in attics or interstitial spaces. Recirculating range hoods or recirculating over-the-range microwaves do not satisfy the kitchen exhaust requirements. For exhaust hood systems capable of exhausting in excess of 400 cubic feet per minute (188 liters per second), provide makeup air at a rate approximately equal to the exhaust air rate. Makeup air systems must have a means of closure and be automatically controlled to start and operate simultaneously with the exhaust system. Use ENERGY STAR-labeled bathroom exhaust fans in all bathrooms (including half-baths) or performance equivalent for projects outside the U.S. A heat recovery ventilator (HRV) or energy recovery ventilator (ERV) may be used to exhaust single or multiple bathrooms if it has an efficacy level meeting the ENERGY STAR Technical Specifications for Residential Heat-Recovery Ventilators and Energy-Recovery Ventilators (H/ERVs) Version 2.0 as certified by the Home Ventilating Institute (HVI).
- Unvented combustion appliances (ovens and ranges excluded) are not allowed.

- ▶ Carbon monoxide (CO) monitor must be installed on each floor of each dwelling unit, hard-wired with a battery backup. CO monitors are required in all types of units, regardless of the type of equipment installed in the unit.
- ▶ Any indoor fireplaces and woodstoves must have solid glass enclosures or doors that seal when closed.
- ▶ Any indoor fireplaces and woodstoves that are not closed combustion or power-vented must pass a backdraft potential test to ensure that depressurization of the combustion appliance zone is less than 5 Pa.
- ▶ Space- and water-heating equipment that involves combustion must be designed and installed with closed combustion (i.e., sealed supply air and exhaust ducting) or with power-vented exhaust, or located in a detached utility building or open-air facility.

HEALTHCARE

Meet the requirements of ASHRAE Standard 170-2017, Sections 6-10, and meet the requirements above for monitoring for mechanical ventilation systems.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The ASHRAE standards have been updated to newer versions:

- ▶ ASHRAE 62.1-2010-> ASHRAE 62.1-2016 (updated version)
- ▶ ASHRAE 170-2008 and FGI 2010 -> ASHRAE 170-2017 (updated version)

Option 2 is available for use through regional compliance pathways.

- ▶ EN 13779-2007 ->EN 16798-3:2017 (local applicability)

Monitoring requirements for smaller ventilation systems have been removed while revising requirements for larger systems per ASHRAE 189.1 and previous LEED interpretation. Specific criteria are introduced for constant-volume systems such as an indicator that confirms damper position to maintain design minimum outdoor airflow during system test and balance.

Credit language for natural ventilation pathway is revised to incorporate ASHRAE 62.1-2016 addendum I and to reflect better which specific sections from 62.1-2016 need to be met for naturally ventilated spaces, for engineered natural ventilation or for ventilation of projects registered as historic buildings (since existing historic buildings may be limited in their ability to alter the building envelope to include additional ventilation systems/adjust openings).

Step-by-Step Guidance

Refer to LEED v4 reference guide with the following modifications:

Step 1. Evaluate Outdoor Air Quality: For evaluation of outdoor air quality, follow the specific sections 4.5, 6.2, 6.5 and 7 per the latest version of ASHRAE 62.1-2016.

Steps 2-4. Refer to LEED v4 reference guide

Mechanically ventilated spaces (and Mixed-Mode spaces when a mechanical ventilation is active)

Steps 1. Refer to LEED v4 reference guide

Step 2-3. Refer to the LEED v4 reference guide. Note: The indoor air quality procedure (defined in ASHRAE Standard 62.1-2016, Section 6.3) may not be used to comply with this prerequisite. For projects interested in using this section must pursue the pilot credit Performance-based indoor air quality design and assessment.

Step 4. Meet Minimum Requirements: Meet minimum requirements of the newer versions of the standards as defined in rating system requirements, for example ASHRAE 62.1-2016, sections 4, 5, 6.2, 6.5 and 7.

For projects within U.S., indicate if project is in a non-attainment area for PM2.5 and ozone. Projects located outside the U.S. are considered to be in non-attainment area for PM2.5 and ozone (exceeding national standard or guideline, unless local government published outdoor air quality data indicates otherwise). For all of these projects, follow additional guidelines in ASHRAE 62.1- 2016, Section 6.2.1.2 and section 6.2.1.3 for exceedance of PM2.5 and ozone. Prior to occupancy, for PM2.5, confirm that prior to occupancy, air cleaning devices or filters with a minimum MERV of at least 11 or higher per ASHRAE 52.2 (or equivalent) are installed. Similarly, for ozone, confirm that prior to occupancy, air cleaning devices will be provided where the ozone levels exceed the NAAQS levels for projects within U.S. or exceed ozone national standard or guideline for projects located outside of the U.S.

Step 5. Implement Airflow Monitoring: Airflow monitoring requirements for mechanically ventilated spaces need to be met only for larger ventilation systems with airflow > 1000 cfm (472 L/s). For constant volume systems, at the time of system test and balance, install a monitoring device or indicator that confirms position of intake damper to maintain the design minimum outdoor airflow.

Naturally ventilated spaces (and mixed mode systems when mechanical ventilation is inactivated)

Steps 1-2. Refer to LEED v4 reference guide

Step 3. Perform Natural Ventilation Procedure: Determine the best of the three options provided to pursue for natural ventilation, depending on if the building is registered as an historic building or not and follow the requirements for either case. Section 6.4.1.6.2 Sizing Openings—Path A is required for natural ventilation openings; Path B is not equivalent and should not be used. For non-historic buildings, exception to full natural ventilation requirements per ASHRAE 62.1-2016 is permitted under section 6.4 for engineered natural ventilation system.

Step 4. Refer to LEED v4 reference guide

Step 5. Implement Monitoring System: Direct exhaust airflow measurement device strategy is not allowed for projects registered as historic buildings.

Further Explanation

Required Documentation

Refer to LEED v4 reference guide with the following modifications.

- ▶ For Option 1, Option 2 and mixed mode, confirm that project meets minimum requirements of applicable sections of ASHRAE 62.1- 2016.
- ▶ For naturally ventilated and mixed mode ventilated projects, confirmation that projects meets requirements of the applicable sections in ASHRAE 62.1-2016.

Healthcare documentation only:

- ▶ Confirmation that project meets minimum requirements of ASHRAE 170-2017, Sections 6-10

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Minimum Indoor Air Quality: Providing and maintaining proper minimum ventilation in existing buildings helps to establish minimum indoor air quality and is an

important strategy to assessing how the building is performing for the occupants with regards to overall indoor environmental quality via the indoor environmental quality performance prerequisite.